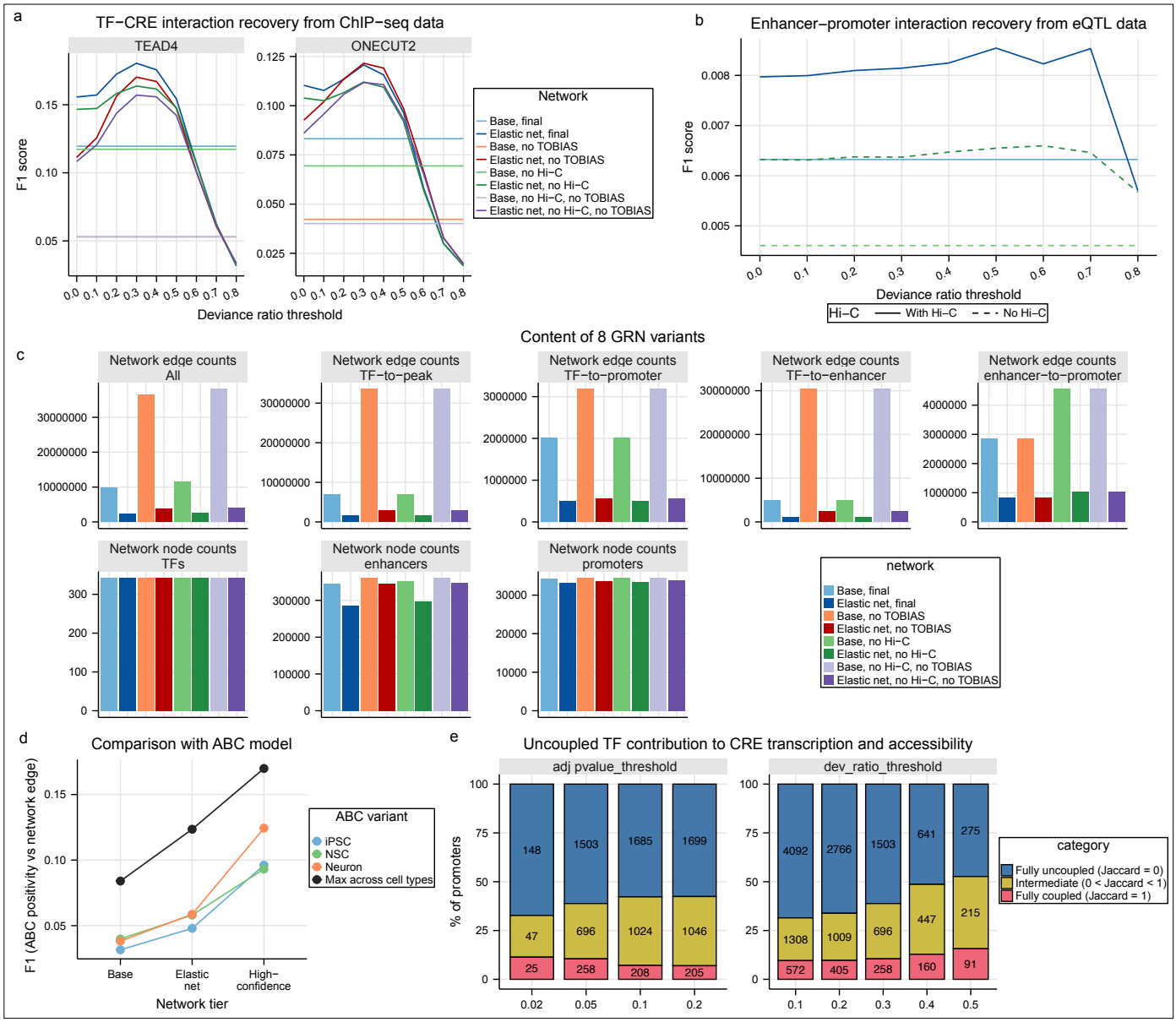


Extended Figure 5 | Validation of GRN filtering thresholds and network composition.

a, F1 scores for recovery of TF-CRE interactions using TEAD4 (left) and ONECUT2 (right) ChIP-seq binding as ground truth, shown across deviance-ratio thresholds for eight GRN variants generated by combinations of inclusion of TOBIAS TF-footprinting information and/or Hi-C chromatin contacts, including their base networks before elastic-net regularization (as horizontal lines). **b**, F1 scores for recovery of enhancer-promoter interactions using genetically supported eQTL enhancer-gene pairs as ground truth, shown across deviance-ratio thresholds for four GRN variants, including "final" and no "Hi-C" with the base networks before elastic-net. **c**, Edge (top) and node (bottom) counts across eight GRN variants. **d**, Concordance between final GRN enhancer-promoter interactions and Activity-by-Contact (ABC) predictions across successive stages of network refinement: base network, Elastic net-selected interactions and the filtered high-confidence network. Concordance is quantified by F1 score using cell-type-specific ABC predictions for iPSC, NSC and neuron, as well as the maximum ABC score across the three cell types. **e**, Overlap between TF sets associated with promoter accessibility and transcription across adjusted p-value (left) and deviance-ratio threshold (right), while adjusted p-value of 0.05 and deviance-ratio of 0.3 are used for the high-confidence GRN. Bar height indicates the number of promoters with at least one TF association in both modalities, and colors indicate whether accessibility- and transcription-associated TF sets are completely distinct, partially overlapping or completely overlapping.